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Studierichting: Informatica
Oefeningenreeks 6G nr. 2.6

$$f(x) = \begin{cases} 0 & \text{als } x \in [-\pi, 0] \\ 2\sin x & \text{als } x \in]0, \pi] \end{cases}$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx$$
$$= \frac{1}{\pi}\int_0^{\pi} 2\sin x\,dx$$
$$= \frac{2}{\pi}\left[-\cos x\right]_0^{\pi}$$
$$= \frac{2}{\pi}(1-(-1))$$
$$= \frac{4}{\pi}$$
$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx$$
$$= \frac{1}{\pi}\int_0^{\pi} 2\sin x\cos(nx)\,dx$$
$$= \frac{1}{\pi}\int_0^{\pi} \left(\sin((1+n)x) + \sin((1-n)x)\right)\,dx$$

Voor $n = 1$:

$$a_1 = \frac{1}{\pi}\int_0^{\pi} \sin(2x)\,dx = 0$$

Voor $n \neq 1$:

$$a_n = \frac{1}{\pi}\left[-\frac{\cos((1+n)x)}{1+n} - \frac{\cos((1-n)x)}{1-n}\right]_0^{\pi}$$
$$= \frac{2((-1)^n + 1)}{\pi(1-n^2)}$$

Dus:

$a_n = 0$ voor oneven n

$$a_{2k} = \frac{4}{\pi(1-4k^2)}$$
$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx$$
$$= \frac{1}{\pi}\int_0^{\pi} 2\sin x\sin(nx)\,dx$$
$$= \frac{1}{\pi}\int_0^{\pi} \left(\cos((1-n)x) - \cos((1+n)x)\right)\,dx$$

Voor $n = 1$:

$$b_1 = \frac{1}{\pi} \int_0^\pi 2 \sin^2 x \, dx$$
$$= \frac{1}{\pi} \int_0^\pi (1 - \cos 2x) \, dx$$
$$= \frac{1}{\pi} \left[x - \frac{\sin 2x}{2} \right]_0^\pi$$
$$= 1$$

Voor $n > 1$:

$$b_n = \frac{1}{\pi} \left[\frac{\sin((1-n)x)}{1-n} - \frac{\sin((1+n)x)}{1+n} \right]_0^\pi$$
$$= 0$$

Fourierreeks:

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$$
$$f(x) \sim \frac{2}{\pi} + \sin x + \sum_{k=1}^{\infty} \frac{4}{\pi(1-4k^2)} \cos(2kx)$$

References